Group Members: Tan Dai Ngo (V01073190), Joyce van de Vegte (V00102221), Jeremy Perkins (V00501015), Da Zhang (V01062902), Naomi Phan (V01097400), Yuanming Wang (V01028720)

Project Description:

Classical and hybrid quantum-classical ML analysis of loan approval dataset

Dataset: https://www.kaggle.com/datasets/sumit12100012/loan-approval-classification/data

This dataset contains 45,000 loan applications with detailed borrower information and loan attributes (13 features), as well as whether the loan was approved (label feature). Exploratory data analysis will form the first phase of the project, including an examination of feature distributions. Statistical analysis will inform feature engineering, including feature transformation and scaling. Methods to address class imbalance will be studied, such as focal loss, weighted cross-entropy, and generating synthetic data (via a generative adversarial network or SMOTE). The data will be modelled in a variety of ways, including SVM, neural networks, XGBoost, random forests, and logistic regression. Performance will be examined for suitable feature sets and hyperparameter settings. Hybrid quantum-classical algorithms will be explored to complement feature transformation (quantum PCA, quantum extreme learning machine), generate synthetic data (quantum Generative Adversarial Networks), and model data (quantum SVM, quantum Neural Network). Performance of quantum-boosted algorithms will be compared to purely classical algorithms. The ultimate goal of the project is to provide analysis perspectives on a loan approval dataset, using a comparison of classical and quantum-classical approaches.